

APP DEV 1 Project Report

1. Student Details

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About Me:

I am a third-year Computer Science undergraduate student specializing in Artificial Intelligence & Machine Learning, with a strong interest in full-stack web development, backend systems, and database design.

2. Project Details

Project Title: Placement Portal Management System

Problem Statement:

To design and build a web-based Placement Portal that allows:

- Students to register , upload resumes, apply for placement drives, and track application status.
- Companies to register, create drives, manage applicants, and update selection status.
- Admin to manage users, approve companies and drives, and monitor placement activities.
- The system must support role-based access control, validation mechanisms, and a structured relational database.

Approach:

The system was developed using Flask and SQLAlchemy with a structured relational database.

Key implementation steps:

- Designed normalized database schema (User, Student, Company, Drive, Application).
- Implemented role-based dashboards (Admin / Student / Company)
- Added authentication with password hashing.

- Implemented drive approval and company approval workflows.
 - Added CGPA eligibility validation and duplicate application prevention.
 - Added deadline expiry check and drive closing logic.
 - Designed responsive UI using Bootstrap
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3. AI/LLM Declaration

I used ChatGPT (GPT-5) for debugging assistance, improving validation logic, and refining documentation.

The extent of AI usage was approximately 15–20%, limited to guidance and formatting support.

All core implementation, database design, testing, and integration were completed independently.

4. Technologies and Frameworks Used

Technology	Purpose
Flask	Backend framework for routing and application logic
SQLAlchemy	ORM for handling database operations
SQLite	Database for storing users, drives, and applications
Jinja2	Template engine for rendering dynamic HTML
Bootstrap 5	Responsive frontend design and styling
Werkzeug Security	Password hashing and authentication
HTML5	Frontend form validation

5. Database Schema / ER Diagram

Tables

The system uses a normalized relational database schema to manage users, drives, and applications efficiently.

User: id, email, password, role, active

Student: id, user_id, name, branch, cgpa, resume

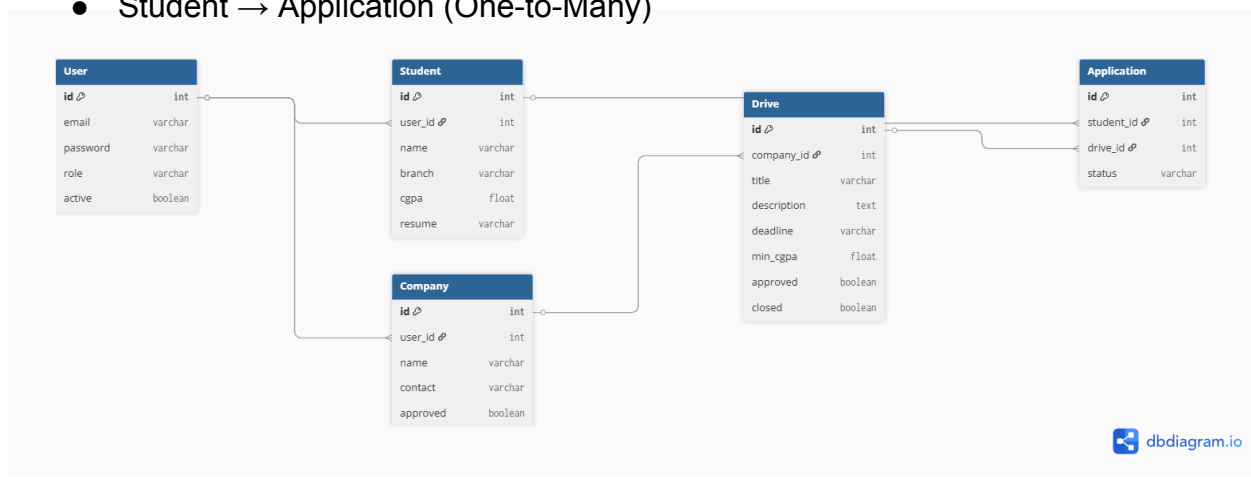
Company: id, user_id, name, contact, approved

Drive: id, company_id, title, description, deadline, min_cgpa, approved, closed

Application: id, student_id, drive_id, status

Relationships

- User → Student (One-to-One)
- User → Company (One-to-One)
- Company → Drive (One-to-Many)
- Drive → Application (One-to-Many)
- Student → Application (One-to-Many)



6. API Resource Endpoints

Endpoint	Method	Description
/login	POST	Authenticate user
/register/student	POST	Register new student
/register/company	POST	Register new Company
/student/dashboard	GET	View student dashboard
/student/drives	GET	View available approved drives
/student/apply/<drive_id>	GET	Apply for a drive
/student/applications	GET	View applied drives and status

/student/upload-resume	POST	Upload resume
/company/dashboard	GET	View company dashboard
/company/create-drive	POST	Create new placement drive
/company/applicants/<drive_id>	GET	View applicants for a drive
/company/update-status/<app_id>/<status>	GET	Update application status
/company/close-drive/<drive_id>	GET	Close a drive
/admin/dashboard	GET	View admin dashboard
/admin/approve-company/<company_id>	GET	Approve Company registration
/admin/reject-company/<company_id>	GET	Reject company registration
/admin/approve-drive/<drive_id>	GET	Approve drive
/admin/reject-drive/<drive_id>	GET	Reject Drive
/admin/toggle-user/<user_id>	GET	Activate / Deactivate user
/admin/users	GET	View all users

7. Video Presentation

Drive Link:

 Placemet_portal_video.mp4